

# The ICT Lounge



Year 7

Year 8

IGCSE

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## Section 7.6: Stage 4 - Implementing a System

After testing completed and any errors/problems have been fixed, you should have a **fully-functional** system.

At this stage, the system is ready to be implemented (**installed**) into the customer's business or place of work.

There are **several options** available as to how this can occur and we will discuss them all in this section.

### Key Concepts of this section:

- # Understand what is meant by **System Implementation**.
- # Know the **different methods** in which new systems can be implemented into an organisation.
- # Be able to **describe advantages** and **disadvantages** of each method.

## Stage 4 - Implementation

### What is system implementation?

- # Implementation just means to '**install**' the system ready for use.
- # **Before** the new system can be installed, there are a number of **tasks** that must be completed.

These are outlined in the table below:

#### Staff training



Staff have to be able to use the new system **before** it is fully installed.

This ensures that the new system can be used straight after it is installed but involves **expensive** and **time-consuming training**.

#### Transfer files from existing system to the new system



If the old system was **paper-based**, existing **paper files** will need to be **scanned** and then **input** into the new computer system.

If the old system was **electronic-based**, existing files will need to be either:

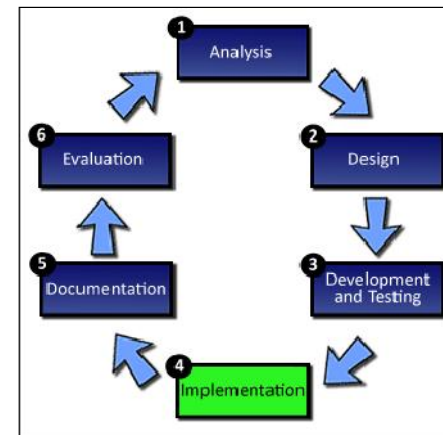
- **Typed into the new system's database manually**
- **Downloaded into the new system.**



### Key Words:

Implementation, Direct, Parallel, Pilot, Phased.

### Examples:



Implementation takes place during the fourth stage of the systems lifecycle.

## Changing over to the new system

- # Once staff are fully trained in the use of the new system and all of the company's files have been transferred, it is time to **changeover** from the **old system** to the **new one**.

Changeover simply means to **switch** from the old system to the new system and this can happen in **four** different ways:

- [Direct changeover](#)
- [Parallel running](#)
- [Pilot running](#)
- [Phased implementation](#)

Each of the changeover methods have **advantages** and **disadvantages** and we will discuss them in detail below:



Implementing a system means to 'install' it.



Staff need to be trained in using the new system.

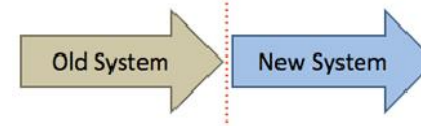
## 1 Direct Changeover

- # With direct changeover, the old system **stops** getting **used one day** and the new system **starts** being **used the next**.
- # The changeover is **very fast** and that's where the name '**direct**' comes from.

### Advantages / Disadvantages of Direct Changeover:

| Advantages                                                                | Disadvantages                                                                                                                                                                                         |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The <b>fastest</b> way of changing from one system to another.            | Staff have to be <b>trained</b> and fully ready to use the new system <b>before</b> it is installed.<br><br>This training can be <b>hard to fit</b> into their <b>daily schedules</b> .               |
| The new system can be used <b>immediately</b> after installation.         | All of the <b>old system's data</b> needs to be uploaded into the new system's <b>database before</b> it is installed.<br><br>Again, this can be <b>time-consuming</b> and difficult to <b>plan</b> . |
| <b>Cheap</b> method of changeover as you don't pay for two sets of staff. | The old system is <b>completely removed</b> so there is nothing to go back to if the new                                                                                                              |

### Examples:



The direct changeover method offers a quick but high-risk way of implementing a new system.  
(click image to zoom)



The direct changeover method is the fastest way of switching from one system to another.

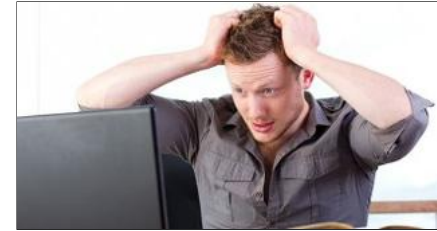
(Only pay the staff who are using the new system)

**Less chance** of the new system being **faulty** as it will already have been fully tested and checked.

system fails.

**Data** can be **lost permanently**.

This can be disastrous and could force the company to shut down for long periods until the system is fixed.



This changeover method is risky because if the new system fails, there is no fallback.

## 2 Parallel Running

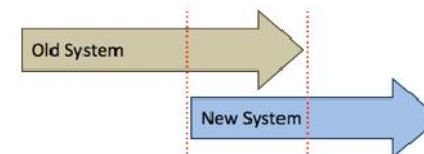
- # With parallel running, the **old system** and the **new system** are run **alongside** each other for period of time.
- # **Data** would be **input** into both the **old system** and the **new system**, producing **two sets** of **results**.

**For Example -** In a **billing system**, both the old system and the new system would produce an **invoice**.

- # When staff are sure they can use the new system and that it is **functioning properly**, the **old system** is **switched off**.

At this point, the **new system** will be **used exclusively**.

### Examples:



The parallel running method offers a low-risk but expensive way of implementing a new system.  
(click image to zoom)



In parallel running, both systems run side by side until the new system has been proven to work properly.



The parallel running method is very expensive compared to other methods.

### Advantages / Disadvantages of Parallel Running:

| Advantages                                                                                                     | Disadvantages                                                                          |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| If the new system fails, the old system is <b>still available</b> .                                            | This changeover method is much <b>more expensive</b> .                                 |
| This way, <b>no data</b> will be <b>lost</b> .                                                                 | This is because you have the <b>running costs</b> of <b>two systems</b> instead of one |
| Staff can be <b>trained</b> in the new system <b>gradually</b> as the changeover doesn't happen immediately.   | (Extra staff to run both systems, electricity etc.)                                    |
| Excellent way to <b>test</b> the <b>new system</b> using <b>live data</b> .                                    |                                                                                        |
| (You can compare the results produced by both systems- any difference would mean an error with the new system) |                                                                                        |

### 3 Pilot Running

- # Implementing a new system using the pilot running changeover method involves introducing the new system to **one part** of the **company** first and then **rolling it out** to **other departments later**.
- # This method is a good way to test that the system works properly without having to massively **restructure** the company.
- # The term '**pilot**' can be defined as '**to experiment or test before introducing something more widely**'.

#### For Example:

If a company had **three** departments - **warehouse**, **payroll** and **sales**, the new system could be installed in the **warehouse** branch **first**.

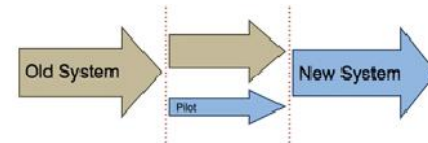
The **old system** would still be used in the **payroll** and **sales** departments.

If the new system was a **success** in the warehouse department, it would be installed into the other two branches later.

#### Advantages / Disadvantages of Pilot Running:

| Advantages                                                                                                                                                             | Disadvantages                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| If the new system <b>doesn't work</b> correctly, it only affects <b>one</b> part of the company.                                                                       | It takes <b>longer</b> to implement the new system across the whole company.<br><br>(This is because the changeover happens gradually from department to department) |
| The implementation is on a <b>smaller scale</b> and is <b>easier to manage</b> than direct changeover.                                                                 | The <b>department</b> taking part in the pilot can <b>lose data</b> if the new system fails.<br><br>(This is because the new system has replaced the old system)     |
| Staff can be <b>trained gradually</b> from department to department.<br><br>The staff that were involved with the pilot scheme can also <b>help train</b> other staff. |                                                                                                                                                                      |
| It is <b>cheaper</b> than parallel running because the pilot is only used in <b>one part</b> of the company.                                                           |                                                                                                                                                                      |

#### Examples:



This method allows the new system to be tested in one department before being installed in others.  
(click image to zoom)



The system could be installed in the warehouse department first.



Once proven to work correctly in the warehouse, the system could be installed into sales and then the rest of the company.



Pilot running takes a long time to fully implement the new system.

## 4 Phased Implementation

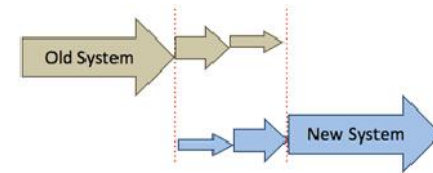
- # This type of changeover is where the new system is implemented **one part at a time**, in **phases**.
- # When each part of the new system has been fully tested and is known to work correctly, the **next part** is **phased in**.
- # Eventually the **old system** will be fully **replaced**.

Sometimes replacing an old system in this fashion is known as '**phasing it out**'.

### Advantages / Disadvantages of Phased Implementation:

| Advantages                                                                                                                                                                                | Disadvantages                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Staff can be <b>trained gradually</b> as they only need to train in the part of the system that is <b>currently</b> being <b>phased in</b> .                                              | More <b>expensive</b> than direct changeover as you have to <b>test</b> each phase <b>separately</b> before moving onto the next.    |
| It is <b>easier</b> to find system <b>errors</b> as you are dealing with only one part at a time.<br><br>(If the whole system is implemented at once, it is harder to pinpoint the error) | Because the system is installed in separate phases, it can take a <b>long time</b> to implement the whole system.                    |
| If one part of the new system <b>fails</b> , it is still possible to access the parts of the old system that are <b>still in use</b> .                                                    | Only suitable for systems that can be <b>split</b> into <b>separate parts</b> .<br><br>(Not all systems can be implemented this way) |

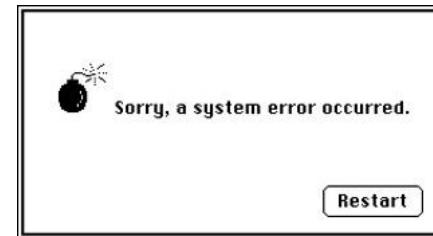
### Examples:



The phased implementation method offers a medium-risk but lengthy method of switching to a new system.  
(click image to zoom)



In this method, the new system is 'phased in' and the old system is 'phased out'.



System errors are easier to weed out because you are only testing one part of the system at a time.

### Which method of implementation is the most suitable?

- # The most suitable method of system implementation depends on the **situation** and the **needs** of the **organisation**.

The table below highlights some examples of **when** each implementation method would be the best choice:

| Implementation method | Suitable situation to use each method |
|-----------------------|---------------------------------------|
|                       |                                       |



### Key Words:

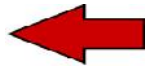
Quick, Data Loss, Afford.

### Activity!

|                               |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Direct changeover.</b>     |   | Best where you want a <b>quick changeover</b> .<br>You simply stop using the old system one day and start using the new one the next.                                                                                                                                                                                                                             |
| <b>Parallel running.</b>      |  | Best for use in departments within organisation's where you <b>cannot afford to lose</b> any <b>data</b> .<br><br>The two systems are run side by side so if there are any problems with the new system, you have the old system to fall back on.<br><br>The organisation must be able to afford the <b>high costs</b> of this type of implementation.            |
| <b>Pilot running.</b>         |  | Best for organisation's that have <b>large, important departments</b> that <b>must not lose</b> any <b>data</b> .<br><br>With pilot running you can use the new system in <b>smaller departments first</b> where any data loss will not be as catastrophic. Once the system is proven to <b>run flawlessly</b> , it can be installed into the larger departments. |
| <b>Phased implementation.</b> |  | Best for organisation's that cannot afford to <b>lose data</b> and can afford the <b>time</b> and <b>cost</b> of a <b>slow changeover</b> .                                                                                                                                                                                                                       |

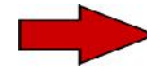
[The Implementation Stage - Research Sheet](#)

Click the above task and answer the questions about the Implementation Stage.



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